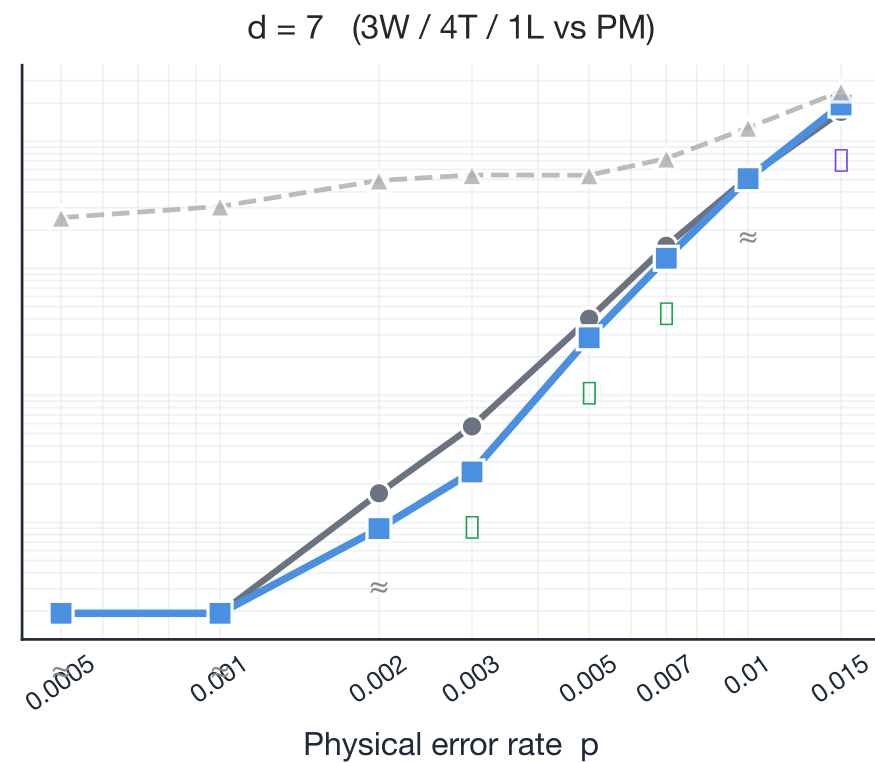
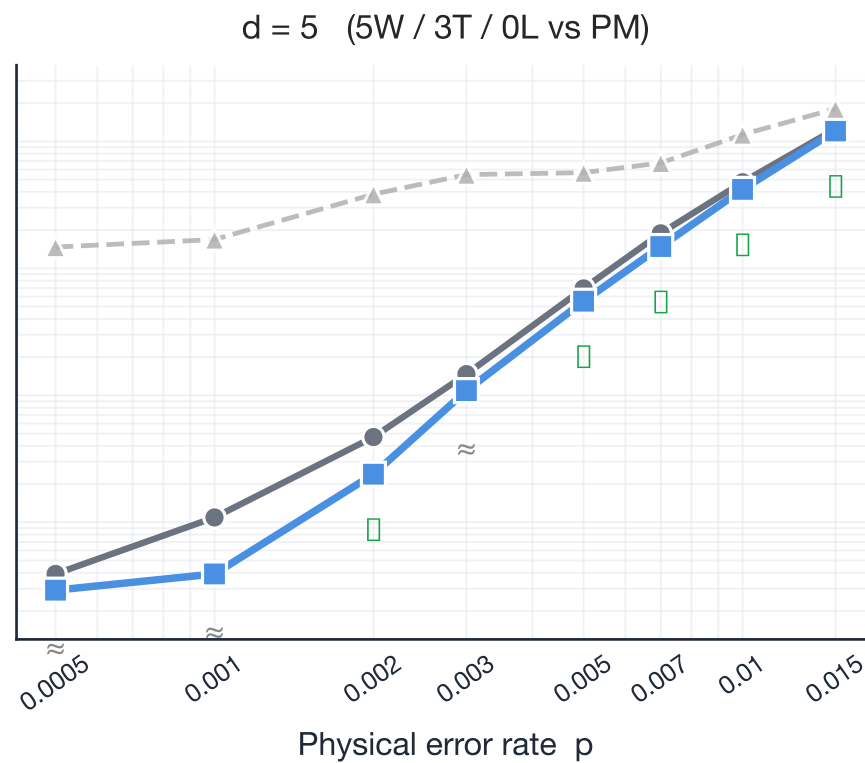
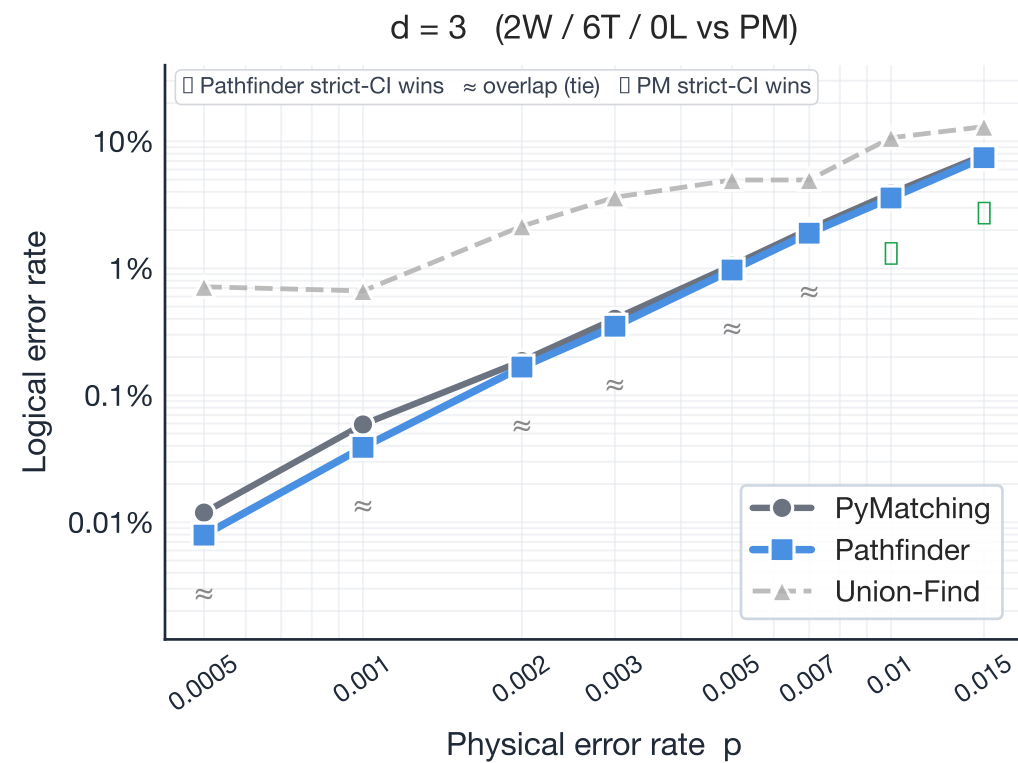


Pathfinder vs PyMatching at $d \in \{3,5,7\}$: 10 strict-CI wins / 13 statistical ties / 1 strict-CI loss (3-param noise, 100K shots/point)



Strict-CI test uses 95% Wilson intervals. Pathfinder's only strict-CI loss is at $d=7, p=0.015$ — the above-threshold regime where the surface code can no longer suppress errors and PM's combinatorial structure is provably near-optimal (same physics as the $d=9, p=0.015$ result of Fig 11).